GENERATIVE AI COURSE

Module 3 — Topic 1
Introduction to Generative AI Tools

Study Notes • Examples • Practical Activity • Quiz

Learning objectives
Understand Generative AI tools, compare AI tools with traditional software, identify popular applications, and choose a suitable tool for a task.

1. What Are Generative AI Tools?

Generative Artificial Intelligence (Generative AI) tools are software applications that use AI models to create new content from user input, instructions, or prompts. Depending on the system, they can generate text, images, presentations, computer code, audio, video, and summaries.

Example
Input: “Explain Artificial Intelligence in simple English.”
AI processing: The model interprets the prompt and generates a response.
Output: A clear explanation of Artificial Intelligence.

Basic workflow

INPUT
User prompt, question, image, or file

AI MODEL
Processes the input and generates content

OUTPUT
Text, image, code, audio, or other content

2. AI Tools vs Traditional Software

Feature	Traditional software	Generative AI tools
Working method	Programmed rules and logic	Trained models generate responses
Input	Commands, forms, structured data	Prompts, questions, files, and sometimes media
Output	Often predefined or rule-based	Newly generated content
Flexibility	Often works within predefined workflows	Can respond to many kinds of instructions
Example	Calculator	ChatGPT
Common use	Calculations, records, transactions	Writing, summaries, ideas, coding assistance

Example: A calculator computes 25 × 4 and returns 100. A Generative AI tool can also explain multiplication and create practice questions. Traditional software can be sophisticated too; Generative AI is not automatically better or more accurate for every task.

3. Popular Generative AI Tools

1. ChatGPT

Writing, learning, brainstorming, supported file analysis, image generation where available, and programming assistance.

https://chatgpt.com/

2. Google Gemini

Questions, writing, learning, multimodal tasks, and supported Google ecosystem workflows.

https://gemini.google.com/

3. Microsoft Copilot

AI-assisted research, writing, and productivity. Microsoft 365 features vary by product and subscription.

https://copilot.microsoft.com/

4. Claude

Drafting, summarizing documents, reasoning through problems, and coding assistance.

https://claude.ai/

5. Canva AI

Visual design workflows, presentations, posters, social graphics, and AI-assisted content creation.

https://www.canva.com/

6. GitHub Copilot

Code suggestions, programming explanations, test generation, and development assistance.

https://github.com/features/copilot

4. Popular Applications of Generative AI

Application	Example task	Expected output
Education	Explain a C++ loop	Explanation and examples
Writing	Draft a formal email	Email draft
Image design	Create a Navratri poster	Poster image
Presentations	Prepare an AI project presentation	Slides
Programming	Write a Python calculator	Source code
Business	Draft a marketing plan	Business document
Research	Summarize reliable source material	Research summary
Productivity	Organize meeting notes	Action-item list
Cybersecurity	Create a phishing-awareness lesson	Safety checklist

5. Choosing the Right AI Tool

Choose a tool according to the task, required output, privacy needs, cost, and available features.

Education and learning: ChatGPT or Google Gemini — explanations, practice questions, and study plans.

Writing and summarization: ChatGPT or Claude — drafting, rewriting, summarizing, and organizing content.

Posters and visual design: Canva AI — posters, presentation layouts, and social graphics.

Programming: GitHub Copilot or ChatGPT — code suggestions, explanations, debugging assistance, and tests.

Office productivity: Microsoft Copilot — supported workflows in Microsoft's ecosystem.

Research: ChatGPT, Gemini, or Claude — organizing research and summarizing information; verify claims against reliable sources.

Five selection factors

1. Task: What should the tool accomplish?
2. Output format: Text, image, presentation, code, audio, or video?
3. Accuracy: Can you verify the answer?
4. Privacy: Is the information appropriate to share?
5. Cost and access: Does the feature fit your budget and subscription?

6. Practical Activity

Project: Create a mini-project called “Introduction to Generative AI.”

> Prompt: “Create a beginner-friendly introduction to Generative AI. Explain its meaning, compare AI tools with traditional software, describe five real-world applications, and recommend suitable tools for education, design, business, and programming. Use simple English, headings, examples, and a summary table.”

Expected output: A clear explanation, comparison table, five applications, tool recommendations, and a short conclusion.

7. Responsible AI

Accuracy: Verify important claims using reliable sources.
Privacy: Avoid uploading passwords, confidential files, or sensitive personal information unnecessarily.
Bias: Check whether the output is unfair or incomplete.
Copyright: Respect the rights and licenses of content.
Security: Test generated code; do not run untrusted code blindly.
Transparency: Disclose AI assistance when required.
Human review: Use appropriate expert review for high-impact decisions.

8. Knowledge Check

1. What is a Generative AI tool?

A. A tool that only stores files
B. A tool that can generate new content
C. A tool used only for calculations

Answer: B

2. Which tool is useful for AI-assisted visual design?

A. Canva AI
B. A basic calculator
C. A file manager

Answer: A

3. Which tool helps developers with code suggestions?

A. GitHub Copilot
B. A photo gallery
C. A music player

Answer: A

4. What should you do with important AI-generated facts?

A. Trust every answer
B. Verify them using reliable sources
C. Ignore the evidence

Answer: B

5. What is important when choosing an AI tool?

A. Task requirements and privacy
B. Logo colour only
C. Number of letters in its name

Answer: A

9. Learning Outcome

After completing this topic, you should be able to define Generative AI tools, explain how they differ from traditional software, identify popular applications, and choose a suitable AI tool for a specific task.

Key takeaway: Select the AI tool that fits your task, verify important outputs, and use AI responsibly.